

Two-level PMU-based Linear State Estimator

Tao Yang, *Student Member, IEEE*, Hongbin Sun, *Member, IEEE*, and Anjan Bose, *Fellow, IEEE*

Abstract—The State Estimator function in a control center today is a suite of three programs solved sequentially: Topology Processing, State Estimation, and Bad Data Detection-Identification. The state estimation equations are nonlinear because the inputs are mostly real and reactive power measurements. A linear state estimator is possible if the inputs are only complex currents and voltages and if there are enough such measurements to meet observability and redundancy requirements. The main contribution in this paper is the suggestion that the Topology Processing function and the Bad Data Detection-Identification be done at each substation rather than at the control center. It is shown how this two-level processing is faster and more accurate leaving the control center level state estimator solution free of the bad data errors that are major problems today.

Index Terms—PMU, Linear State Estimation, Distributed Database.

I. INTRODUCTION

The traditional state estimator function in a control center is actually made up of three programs solved sequentially:

1. Topology Processor (TP) that uses the real-time circuit breaker status with the substation and system level topology to determine the connectivity of the whole network;
2. State Estimation (SE) that solves for the complex voltages at each bus from the real-time analog measurements;
3. Bad Data Detection-Identification (BD) which tests the solution to find bad measurements (and if found, reruns the SE solution without the bad data).

Topology errors are a major issue for the state estimators in use today [1]-[3]. First, they cause very large errors in the SE solution which in turn makes the downstream applications (contingency analysis, optimal power flow, nodal prices,

billing reconciliation, etc.) erroneous. Second, topology errors are more difficult to detect and identify than analog measurement errors [4]. Third, topology processing is dependent on the circuit breaker status measurements which have no redundancy (unlike the analog measurements).

The state estimation equations are nonlinear because most of the analog measurements are real and reactive powers, so the solution requires iterating to convergence [5]. Convergence is hindered by bad data and particularly so if the error is in the topology. If all the analog measurements were synchronized currents and voltages, then the state estimation equations would be linear, thus eliminating the convergence issue. Accuracy will still be dependent on the measurement error distributions, thus redundancy will still be required. To have enough phasor measurements to meet observability and redundancy requirements would mean much higher penetrations of PMU technologies than is available today. However, some regions (like in China) are already aiming for this in the next few years while more will attain this if the very large numbers of intelligent electronic devices (IED) at high voltage substations are fitted with time synchronization signals.

In this paper, we propose a distributed two level state estimator function that uses a linear SE algorithm at the control center level but moves the local topology processing and the bad data detection-identification to the substation level. The local topology processing and bad data detection-identification at each substation is done with very different algorithms than is used today at the control center level. In fact, substation level topology processing and bad data detection-identification are no longer separate functions but analog measurement errors and circuit breaker status errors are identified together. The result is that both the topology results and the measurement data are filtered locally to provide a much more accurate set of data to the control center level state estimation calculation. Unlike the present state estimator and some other distributed state estimators [6]-[10], the bad data detection-identification and filtering is done prior to the state estimation solution rather than after. The linearity of the state estimator guarantees a solution and the substation level pre-filtering of topology and measurement errors guarantee the accuracy of the solution.

In the next section, the linear state estimation formulation is described. In section III, we describe the new substation level calculations that pre-filter all the real-time data at that substation; we call this the Substation Level State Estimator or Zero Impedance State Estimator. In section IV, we briefly describe the communications and distributed database

This work was supported in part through the Consortium for Electric Reliability Technology Solutions (CERTS), funded by the Assistant Secretary of Energy Efficiency and Renewable Energy, Office of Distributed Energy and Electricity Reliability, Transmission Reliability Program of the U.S. Department of Energy under Interagency Agreement No. DE-AI-99EE35075 with the NSF. This work was also supported by the Power System Engineering Research Center (PSERC).

Tao Yang is with Washington State University, Pullman, WA 99163 USA (e-mail: tyang2@eecs.wsu.edu).

Hongbin Sun is with State Key Lab of Power Systems, Department of Electrical Engineering, Tsinghua University, Beijing 100084, China. (e-mail: shb@tsinghua.edu.cn).

Anjan Bose is with Washington State University, Pullman, WA 99163 USA (e-mail: bose@wsu.edu).

architecture that will be required to support such a two-level state estimator. Finally, we show some experimental results for the substation level calculations in section V.

II. CONTROL CENTER LEVEL LINEAR STATE ESTIMATOR

If all the analog measurements were synchronized currents and voltages, then the state estimation equations would be linear. With the help of increasing installations of phasor measurements, we can implement a new measurement function of the state estimator which is linear in the complex plane. In this state estimator, both the states and the measurements are defined in the complex plane and as with the traditional state estimator, the power system states are the complex bus voltages.

In general, state estimation determines the most likely state of the system based on the quantities that are measured. Here we use the maximum likelihood estimation (MLE) method which is popularly used. So the linear state estimation is the following optimization problem:

$$\begin{aligned} \text{Min } \tilde{\mathbf{r}}^T \tilde{\mathbf{W}} \tilde{\mathbf{r}} \\ \text{s.t. } \tilde{\mathbf{z}} = \tilde{\mathbf{H}} \tilde{\mathbf{x}} + \tilde{\mathbf{r}} \end{aligned} \quad (1)$$

where $\tilde{\mathbf{r}}$ is the residues vector, $\tilde{\mathbf{W}}$ is the weight matrix, $\tilde{\mathbf{z}}$ is the measurements vector, $\tilde{\mathbf{x}}$ is the vector of system states, and $\tilde{\mathbf{H}}$ is the measurement function matrix relating the measurement vector to the states.

To the control center level state estimator, given a power system with n buses and m branches, the measurement inputs are the bus voltages $\tilde{\mathbf{V}}_{bus} = [\tilde{v}_{bus,1}, \dots, \tilde{v}_{bus,n}]^T$, branch currents at both ends $\tilde{\mathbf{I}}_{b1} = [\tilde{i}_{b,1}, \dots, \tilde{i}_{b,m}]^T$, $\tilde{\mathbf{I}}_{b2} = [\tilde{i}_{b,m+1}, \dots, \tilde{i}_{b,2m}]^T$, and the injection currents $\tilde{\mathbf{I}}_{inj} = [\tilde{i}_{inj,1}, \dots, \tilde{i}_{inj,m}]^T$. The states of the system are the bus complex voltages $\tilde{\mathbf{x}} = [\tilde{v}_1, \dots, \tilde{v}_n]^T$. Assuming the system admittance matrix is $\tilde{\mathbf{Y}}$, the measurement function should be:

$$\tilde{\mathbf{z}} = \begin{pmatrix} \tilde{\mathbf{V}}_{bus} \\ \tilde{\mathbf{I}}_{b1} \\ \tilde{\mathbf{I}}_{b2} \\ \tilde{\mathbf{I}}_{inj} \end{pmatrix} = \tilde{\mathbf{H}} \tilde{\mathbf{x}} + \tilde{\mathbf{r}} = \begin{pmatrix} \mathbf{I} \\ \tilde{\mathbf{Y}}_1^b \\ \tilde{\mathbf{Y}}_2^b \\ \tilde{\mathbf{Y}} \end{pmatrix} \tilde{\mathbf{x}} + \tilde{\mathbf{r}} \quad (2)$$

where $\tilde{\mathbf{Y}}_1^b$, $\tilde{\mathbf{Y}}_2^b$ can be derived from the branch admittances.

Then we can get the solution of the state estimation problem by the usual formulation:

$$\tilde{\mathbf{x}} = (\tilde{\mathbf{H}}^T \tilde{\mathbf{W}} \tilde{\mathbf{H}})^{-1} \tilde{\mathbf{H}}^T \tilde{\mathbf{W}} \tilde{\mathbf{z}} \quad (3)$$

In the linear state estimator, the cost function is still the weighted least squares (WLS) problem while the measurement function is linear. Compared to the traditional state estimator, our control center level state estimator does not require iterations during the process of state estimating, thus not suffering from divergence and needing much less

calculation time. We can see from the algorithm that we need all the substations to provide synchronized voltage and current measurements to guarantee observability. This requirement is similar to that of the traditional state estimator and so is the requirement for redundancy.

III. SUBSTATION LEVEL LINEAR STATE ESTIMATOR

With more and more sensors installed in substations, the substation measurements become more and more abundant to enable processing bad data of both analog measurements and circuit breaker statuses at the substation level. Previous papers have focused on the processing of substation measurements and the SuperCalibrator method [10] has also been tested in the field. But these methods all use the traditional topology processing, state estimation and bad data processing within the substation as used at the control center level and suffer from some of the same difficulties.

The substation level state estimator we propose here uses all the current measurements including those through the circuit breakers to estimate both the substation current states as well as the substation topology without having to use the circuit breaker statuses first based on the circuit breaker oriented substation model shown in Fig 1. Instead, the topology obtained from this processing is then compared to the breaker statuses to verify their accuracy.

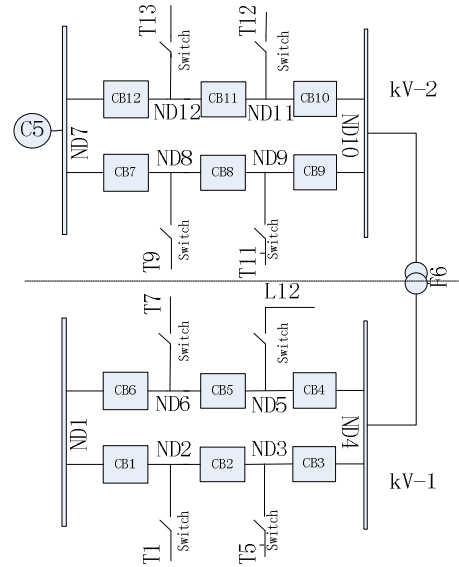


Fig 1. Circuit Breaker Oriented Substation Model

The main function of the substation level state estimator proposed here is to use all the digital and analog measurements in the substation to estimate the real time digital states and analog phasor data of bus voltages, branch currents, and injection currents. As before, the only analog measurements used are the synchronized voltage and current measurements, thus making all the equations linear. This filtered substation data together with the substation topology is provided to the control center as input to the state estimation at that level.

The substation level state estimator proposed here consists of two parts, one processing the current measurements and the

other voltages. As there are no impedances at one voltage level within the substation, these two are called zero impedance current state estimator and zero impedance voltage state estimator, respectively. The brief flow chart of the substation level state estimator is shown in Fig 2.

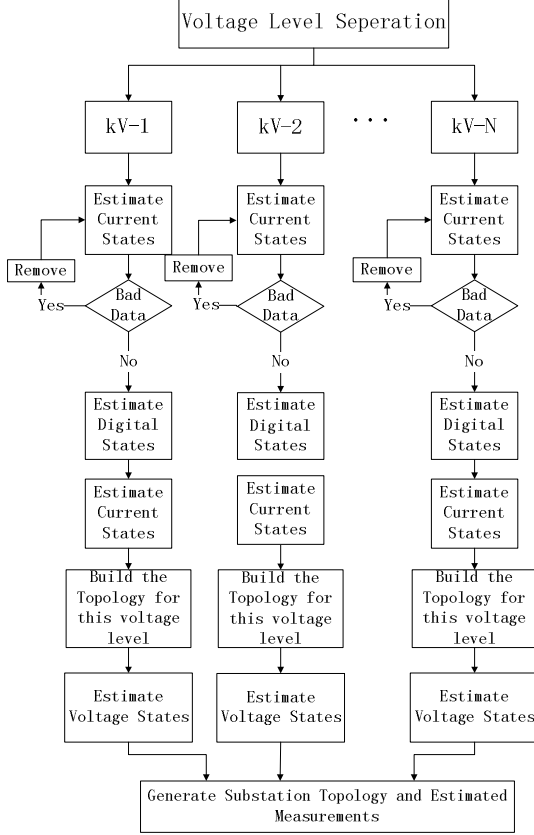


Fig 2. Flow Chat of the Substation Level State Estimator

Each substation is divided into several kV systems such that each kV system is one voltage level. For example, in the substation of Fig 1, there are two kV systems named kV-1 and kV-2 respectively, which are connected by a transformer F6.

In each kV system, as there is no impedance, we cannot find the measurement function relating the voltages and the branch current measurements. Thus we calculate a new state estimator for each kV system to handle the current and voltage measurements separately. The zero impedance current state estimator defines the currents on circuit breakers as states and uses state estimation equations based on Kirchhoff's Current Law (KCL) to estimate the states. The equations for the current measurements at one voltage level, i.e. with no impedance connections, can be written as follows:

$$\tilde{\mathbf{z}}_{inj} = \mathbf{A}_{KCL} \tilde{\mathbf{x}} + \tilde{\mathbf{r}}_{inj} \quad (4)$$

where $\tilde{\mathbf{z}}_{inj}$ is the injection current at each node, $\mathbf{A}_{KCL}$ is the adjacent matrix relating nodes to circuit breakers in the zero impedance power system, $\tilde{\mathbf{x}}$ is the state vector of the circuit breaker currents, and $\tilde{\mathbf{r}}_{inj}$ is the corresponding residue vector.

The equations for the current measurements at each circuit breaker have the obvious relationship:

$$\tilde{\mathbf{z}}_{cb} = \mathbf{I} \tilde{\mathbf{x}} + \tilde{\mathbf{r}}_{cb} \quad (5)$$

where $\tilde{\mathbf{z}}_{cb}$ is the vector of current measurements for each circuit breaker, $\mathbf{I}$ is the identity matrix, and $\tilde{\mathbf{r}}_{cb}$ is the corresponding residue vector.

Then the measurements functions can be represented by:

$$\tilde{\mathbf{z}} = \begin{pmatrix} \tilde{\mathbf{z}}_{inj} \\ \tilde{\mathbf{z}}_{cb} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{KCL} \\ \mathbf{I} \end{pmatrix} \tilde{\mathbf{x}} + \begin{pmatrix} \tilde{\mathbf{r}}_{inj} \\ \tilde{\mathbf{r}}_{cb} \end{pmatrix} = \mathbf{H} \tilde{\mathbf{x}} + \tilde{\mathbf{r}} \quad (6)$$

This is also a linear state estimation problem, so we can find the estimation solution by (3) with the difference that the entries in this H matrix are 1, 0, or -1 and the states are currents. The calculation is simple and fast.

Once the currents of all circuit breakers are estimated, the analog bad data can be identified and rejected by the traditional testing method based on maximum residue. The zero impedance current state estimation is repeatedly executed until no bad data remains.

The final estimated circuit breaker currents can then be directly utilized to verify the digital status of the corresponding circuit breakers to identify any topology errors. For example, if the estimated current for a circuit breaker is not close to zero but the digital measurement of this circuit breaker is open, we conclude that the status measurement is bad and the real state of the circuit breaker is closed. We call this verification process the digital state estimation.

After the bad status data are eliminated, we need to repeat this algorithm again because the new topology will give us a new adjacent matrix $\mathbf{A}_{KCL}$ to enable a more precise estimation of the currents.

We can see from this algorithm that the analog and the digital states are decoupled, i.e. the analog estimation can be done without knowing the digital status correctly as long as there is enough redundancy. The advantage of this is we can identify the bad data of analog and status directly instead of using the hypothesis testing method to find the topology error indirectly as done today.

After we get the estimated digital status of each circuit breaker, we can do topology analysis for this voltage level. The outputs will include how many buses this voltage level has and the relationship between buses and nodes, which can be sent to the control center.

Finally, the voltage at each bus needs to be calculated. In the zero impedance voltage state estimator, we define the state as the voltage of each bus, and the measurements are the voltage phasor measurements at the nodes belonging to the bus. Actually the solution is the weighted average value of all the voltage measurements on that bus.

Although the substation level state estimator proposed here has several steps, the calculations are very simple and fast. At the end, this calculation can output any analog values based on the estimated states to the control center together with the substation topology. At a minimum, we assume that branch currents, bus injection currents, and bus voltages for each

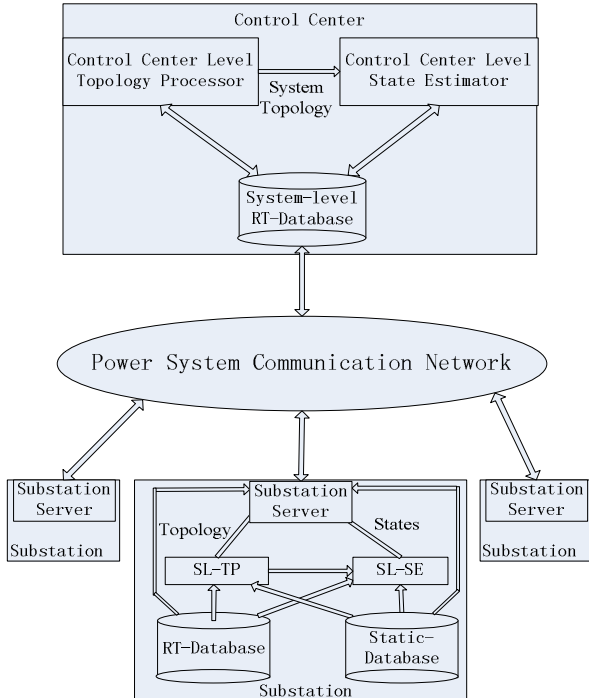
voltage level will be transferred. The main advantages of this substation level state estimator are:

- (1) It utilizes all of the current and voltage measurements in the substation and provides a more accurate phasor measurement set to the upper level applications.
- (2) It provides a direct way to estimate the status of circuit breakers from analog measurements to avoid topology errors.

IV. ARCHITECTURE OF THE DECENTRALIZED TWO-LEVEL STATE ESTIMATOR

This two-level state estimator will require a new architecture for the power system state estimation system including a distributed database and communication system. In the present system, the static database of all substation connectivity information, at the node-circuit breaker level, resides at the control center. The real-time raw data from each substation is periodically transferred to the control center over the RTU to SCADA communication system.

In our proposed scheme, the substation level state estimator needs to estimate the substation state and build the substation level topology, thus requiring the substation connectivity data at the substation. In Fig 3, we show a scheme of a distributed database that can support the two-level state estimator. Although networked communication is assumed in Fig 3 because that is the expected communication scheme of the future [11], the two-level state estimator can be supported by the present star-connected communication as long as it is of high enough bandwidth to carry the extra data.



SL-TP: Substation Level Topology Processor

SL-SE: Substation Level State Estimator

RT: Real Time

Fig 3. Decentralized Real Time Modeling System and Database

At each substation, we assume a Substation Server which will manage all the data, communications and the processing

of state estimation. (Obviously, this server will often be a cluster of distributed CPUs.) We can see from Fig 3, that at each substation the static database of that substation's connectivity must be stored and maintained. Keeping this database updated locally may be an easier task than keeping all substation data updated at the control center. The real-time database at the substation is expected to be larger than the present one in the RTU as the RTUs usually handle a much smaller subset of the available substation measurements.

Such a distributed database architecture, because redundancy can be built in, is more flexible and secure, and significantly improves the operating resilience of the control center functions against physical and cyber attack and natural disasters. Also, database maintenance, which is a major issue today, may become easier when updates are done locally.

V. EXPERIMENTS

We use the IEEE-14 bus system in Fig 5, Appendix A as the test system. In this system, the buses connected by transformers are considered to be in one substation. For each substation, we created the circuit breaker oriented single phase model for each substation. For example, Fig 2 is the node-circuit breaker architecture of substation 5 shown in Fig.5.

We use a steady state power flow condition to generate the real-time measurement sets and consider these the true values. To make these measurements emulate real system measurements, Gaussian White Noise is added to all measurements.

Using the kV-1 system in Fig 2 as the example system, we generated a set of both the digital measurements (Table I) and analog measurements (Table II):

TABLE I
DIGITAL MEASUREMENT SET FOR SUBSTATION LEVEL STATE ESTIMATOR

	True Value (p.u.)	Measurements (p.u.)
CB1 current	0	0
CB2 current	1	1
CB3 current	0	1 (Bad Data)
CB4 current	1	1
CB5 current	1	1
CB6 current	0	0

TABLE II
ANALOGUE MEASUREMENT SET FOR SUBSTATION LEVEL STATE ESTIMATOR

	True Value (p.u.)	Measurements (p.u.)
BS1 injection	-0.0317 - j0.0051	-0.0317 - j0.0051
BS2 injection	-0.6040 + j0.3811	-0.6040 + j0.3811
BS3 injection	0.5929 - j0.3874	0.5929 - j0.3874
BS4 injection	0.4390 + j0.1090	0.4390 + j0.1090
BS5 injection	-0.0826 - j0.4821	-0.0826 - j0.4821
BS6 injection	-0.3342 + j0.3442	-0.3342 + j0.3442
CB1 current	0.0053 + j0.0177	0.0053 + j0.0177
CB2 current	-0.6095 + j0.3609	-0.6095 + j0.3609
CB3 current	0.0025 + j0.0003	0.0725 - j0.1484 (Bad Data)
CB4 current	0.4349 + j0.1225	0.4349 + j0.1225
CB5 current	0.3583 - j0.3566	0.3583 - j0.3566
CB6 current	0.0021 + j0.0006	0.0021 + j0.0006

In addition to the Gaussian white noise, the measurements also include one status bad data and one analog bad data. Circuit breaker 1 & 3 & 6 are open with bad data of current on circuit breaker 3 and bad data of digital measurements on circuit breaker 3.

This mixture of digital and analog bad data is difficult to identify by the traditional state estimator. In contrast, we used the zero impedance state estimation algorithm and formula in (6) to estimate the branch current and bus voltage at the same voltage level (kV-1). We can see from the residues and standard error estimated shown in Table IV, our zero impedance current state estimator can detect and identify the bad data on the CB3 current measurement. After we removed this measurement and re-estimated the states, the results and residues become reasonable. We also see from Table III that the estimated digital status for CB3 is open whereas the measurement was closed thus flagging the bad status.

TABLE III
ESTIMATED STATES OF THE ZERO IMPEDANCE CURRENT STATE ESTIMATOR

States	Analogue	Digital
CB1	-0.0044 + j0.0028	0
CB2	-0.6069 + j0.3776	1
CB3	-0.0099 + j0.0005	0
CB4	0.4332 + j0.1199	1
CB5	0.3531 - j0.3545	1
CB6	0.0161 - j0.0006	0

TABLE IV
RESIDUE ANALYSIS

	Residue (p.u.)	Standard Error Estimated (p.u.)
BS1 injection	0.0053	0.0072
BS2 injection	0.0164	0.0220
BS3 injection	0.0532	0.0717
BS4 injection	0.0442	0.0596
BS5 injection	0.0179	0.0242
BS6 injection	0.0154	0.0208
CB1 current	0.0124	0.0185
CB2 current	0.0384	0.0572
CB3 current	0.0974	0.1452
CB4 current	0.0270	0.0403
CB5 current	0.0068	0.0101
CB6 current	0.0192	0.0287

There are analog and digital bad data together on circuit breaker 3. If we transferred these data to the control center, the traditional state estimator would have both a topology error as well as an analog bad data. From the results in Table III and IV, we can see that the substation level state estimator can detect and identify both errors and provide the control center with corrected inputs.

Table V is the estimated states of the IEEE-14 bus system as calculated by the control center level linear state estimator. The results are very accurate because the substation level filtering has not only eliminated the bad data but has also made some correction to the random noise errors.

TABLE V
RESULTS COMPARISON WITH ONE ANALOGUE BAD DATA

	Estimated States		True States	
	Voltage (p.u.)	Angles (Rad)	Voltage (p.u.)	Angles (Rad)
Bus1	1.0597	0	1.06	0
Bus2	1.0447	-0.0869	1.045	-0.0869
Bus3	1.0100	-0.2220	1.01	-0.222
Bus4	1.0186	-0.1803	1.019	-0.1803
Bus5	1.0196	-0.1533	1.02	-0.1532
Bus6	1.0694	-0.2486	1.07	-0.2482
Bus7	1.0613	-0.2335	1.062	-0.2334
Bus8	1.0892	-0.2333	1.09	-0.2332
Bus9	1.0552	-0.2610	1.056	-0.2608
Bus10	1.0502	-0.2638	1.051	-0.2635
Bus11	1.0562	-0.2585	1.057	-0.2581
Bus12	1.0544	-0.2635	1.055	-0.263
Bus13	1.0494	-0.2651	1.05	-0.2646
Bus14	1.0354	-0.2804	1.036	-0.28

VI. CONCLUSIONS

The new synchrophasor measurement units are expected to bring a revolution in power system applications. We propose in this paper, a decentralized two-level linear state estimator based on these phasor measurements. We first use the zero impedance state estimator to estimate the substation level analog states, digital states, and substation topologies. Then we transfer these filtered or estimated substation phasor measurement data with the topology of the substation to the control center instead of the raw data sent nowadays through the SCADA system. Thus we can use this phasor data and the substation topologies to build the system topology and estimate the power system states linearly.

The advantages of this two-level linear state estimator include:

- (1) A linear solution to the system state that always guarantees a solution.
- (2) Pre-filtering of the substation real-time data at the substation to provide more accurate input to the system level state estimation calculations. Thus, the bad data detection, identification and elimination is done before the state estimation calculation rather than after, guaranteeing the accuracy of the calculated system state.
- (3) The substation level calculations are not only linear but also distributed over many substations, thus making this processing very fast. The topology processing and bad data detection at the substation level are also much simpler from a computational viewpoint.
- (4) As the calculations are distributed between the substations and the control center, the database for the static data used in the calculations are also distributed.. Instead of one very large database at the control center, the substation information can be stored at each substation making the

updating of these substation databases easier.

VII. APPENDIX

A. IEEE 14-bus system in experiment

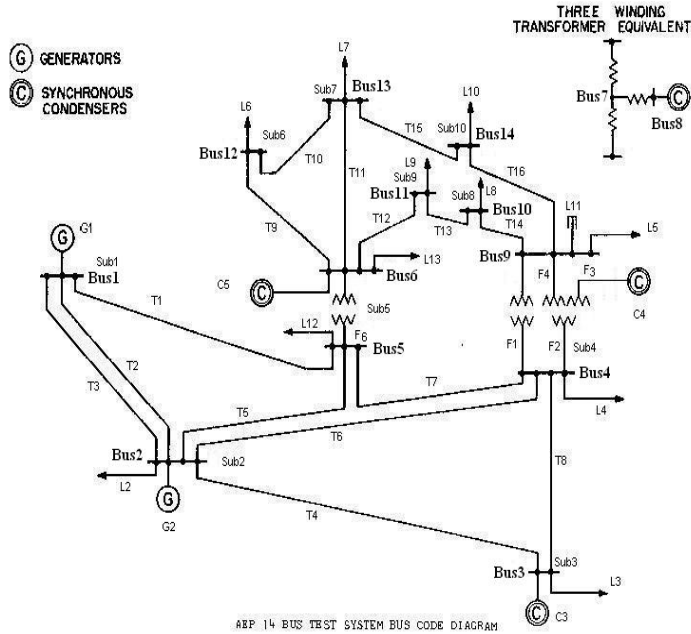


Fig 5. IEEE 14-bus system with named equipments

T-Transmission Line

F-Transformer

G-Generator

C-compensator

VIII. REFERENCES

- [1] R.L. Lugtu, D.F. Hackett, K.C. Liu and D.D. Might, "Power System State Estimation: Detection of Topological Errors", *IEEE Trans. on Power Appar. and Sys.*, vol. PAS-99, No.6, pp.2406-2411, 1980.
- [2] F.F. Wu and W.H. Liu, "Detection of Topological Errors by State Estimation", *IEEE Winter Meeting 1988*, Paper no.216-4.
- [3] A. M. Sasson, S. T. Ehmann, P. Lynch and L. S. Van Slyke, "Automatic, "Power System Network Topology Determination," *IEEE Transactions on Power Apparatus and Systems*, vol. PAS-92, pp. 610-618, March/April 1973.
- [4] M. Bertran and X. Corbella, "On the Validation and Analysis of a New Method for Power Network Connectivity Determination," *IEEE Transactions on Power Apparatus and Systems*, vol. PAS-101, pp. 316-324, Feb. 1982.
- [5] R. A. M. Van Amerongen, "On convergence analysis and convergence enhancement of power system least-squares state estimators," *IEEE Trans. Power Syst.*, vol. 10, no. 4, pp. 2038-2044, Nov. 1995.
- [6] H.B. Sun, B.M. Zhang, Global state estimation for whole transmission and distribution networks, *Electric Power System Research*, May 2005, Vol.74, No.2, pp.187-195.
- [7] D. M. Falcao, F. F. Wu, L. Murphy, "Parallel and distributed state estimation," *IEEE Transactions Proceedings of the 41st Hawaii International Conference on System Sciences - on Power Systems*, vol 10, No 2, pp 724-730, May 1995.
- [8] R. Ebrahimi, R. Baldick, "State estimation distributed processing [for power systems]," *IEEE Transactions on Power Systems*, vol 15, No 4, pp 1240-1246, Nov. 2000.
- [9] M. M. Nordman, M. Lehtonen, "Distributed agent-based state estimation for electrical distribution networks," *IEEE Transactions on Power Systems*, vol 20, No 2, pp 652-658, May 2005.

[10] A. P. Sakis Meliopoulos, George J. Cokkinides, Floyd Galvan, Bruce Fardanesh, "Distributed State Estimator - Advances and Demonstration," *Proceedings of the 41st Hawaii International Conference on System Sciences* - 2008

[11] Carl H. Hauser, David E. Bakken, Ioanna Dionysiou, K. Harald Gjermundrød, Venkata S. Iruva, Joel Helkey and Anjan Bose, "Security, Trust, and QoS in Next-Generation Control and Communication for Large Power Systems," *International Journal of Critical Infrastructures*, 4:1/2, Interscience, 2008,

Tao Yang (StM'2008) was born in Chongqing, P.R.China, in 1983. He received the B.-Eng. degree from the Department of Automation, Tsinghua University in 2005. He is now pursuing his Ph.D. degree in Washington state university. His special fields of interest included power system state estimation, distributed system and real time control.

Hongbin Sun (M'2000) received his double B.S. degrees from Tsinghua University in 1992, the Ph.D from Dept. of E.E., Tsinghua University in 1997. He is now a full professor in Dept. of E.E., Tsinghua Univ, and assistant director of State Key Laboratory of Power Systems in China. From 2007.9 to 2008.9, he was a visiting professor with School of EECS at the Washington State University in Pullman. He is members of IEEE PES CAMS Cascading Failure Task Force and CIGRE C2.13 Task Force on Voltage/Var support in System Operations. His research interests include energy management system, voltage optimization and control, applications of information theory and intelligent technology in power systems.

Anjan Bose (F'89) received the B.Tech. (Honors) from the Indian Institute of Technology, Kharagpur, the M.S. from the University of California, Berkeley, and the Ph.D. from Iowa State University. He has worked for industry, academe and government for 40 years in power system planning, operation and control. He is currently Regents Professor and holds the endowed Distinguished Professor in Power Engineering at Washington State University, Pullman, WA. Dr. Bose is a member of the National Academy of Engineering and the recipient of the Herman Halperin Award and the Millennium Medal from the IEEE.